

HARSHAL BALKRUSHNA MARATHE

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Profile Summary

Entry-Level Data Scientist, Data Analyst with a Master's in Statistics and strong foundation in data analysis, machine learning, and NLP. Experienced in implementing real-world projects involving stock prediction, sentiment analysis, and statistical modeling. Proficient in Python, Adv.Excel, R, SQL, and Data visualization(Power BI and Tableau).

Education

M.Sc. in Statistics - CGPA: 8.73 KBC North Maharashtra University, Jalgaon Relevant Coursework: Regression, Data Mining, Multivariate Analysis, Time Series, Stochastic Processes	Jul 2023 – May 2025
B.Sc. in Statistics – CGPA: 9.54 Moolji Jaitha College, Jalgaon	Jul 2020 – May 2023
HSC (12th) – 73.38% Moolji Jaitha College, Jalgaon	Jul 2018 – Apr 2020

Experience

Data Analyst Intern May 2024 – Jun 2024

Chief Executive Office, Zilla Parishad, Jalgaon

- **Problem statement** - Zilla Parishad needed to understand why some departments failed to meet yearly targets and what led to success in others, based on incomplete RFD data.
- **Results** - Identified key reasons for unmet targets, such as delays in fund allocation, staffing shortages, and operational challenges in specific schemes. Documented successful initiatives, highlighting effective strategies (e.g., community engagement, early planning) that led to target achievement.
- **Impact** - Enabled data-driven decisions and improved planning for future targets by providing clear insights and visual reports to leadership.

Projects

Global Temperature Analysis Using ARIMA and LSTM Models 🔄 Jan 2025 – May 2025

- Analyzed 140 years (1880–2020) of daily global temperature data to identify long-term climate trends. Built and compared ARIMA, SARIMA, and LSTM models for time series forecasting.
- Segmented data into 20-year intervals to study evolving climate patterns over time. Assessed forecasting accuracy across 20-year slots; MAE and RMSE remained stable (MAE: 0.0923–0.1185, RMSE: 0.1169–0.1500), reflecting consistent model performance.
- Compared model performance metrics, where ARIMA achieved lower average MAE (0.1405) and RMSE (0.2442) than LSTM (MAE: 0.2471, RMSE: 0.3160), indicating better forecasting accuracy.

News Sentiment and Stock Movement Prediction – A Statistical NLP Framework 🔄 Sep 2024 – Dec 2024

- Developed NLP-based LSTM model integrating financial news sentiment and stock data.
- Achieved 84% test accuracy and RMSE of 1.25.
- Sentiment integration improved model accuracy by 12% over baseline.

Statistical Models and Applications in NLP 🔄 Jul 2024 – Sep 2024

- Built an end-to-end NLP pipeline for sentiment analysis using Python. Applied Naive Bayes classifier for text classification using scikit-learn.
- Focused on preprocessing, feature extraction, and model evaluation using NLTK and spaCy.

Technical Skills

Programming: Python, Adv. Excel, R, SQL, C/C++, MATLAB, SAS, SPSS

Libraries/Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, NLTK, spaCy

Machine Learning: Regression, Classification, Clustering, Time Series Forecasting, Hypothesis Testing, PCA, Model Evaluation, A/B Testing

Visualization: Tableau, Power BI, Excel, ggplot, Plotly **Big Data & Cloud :** Spark, PySpark, AWS, GCP

Soft Skills: Leadership, Communication, Strategic Thinking, Flexibility

Certifications

NPTEL – Data Analytics in Python Jan – Apr 2024

Score: 83%

UDEMY - Complete Data Analyst Bootcamp May - Jun 2025

Master Data Analysis: Python, Statistics, EDA, Feature Engineering, Power BI, and SQL Server, Snowflake, AWS